



Tribhuvan University

Faculty of Humanities and Social Sciences

BN Organic

E-COMMERCE PROJECT REPORT

Submitted to

Department of Computer Science and Application

Mechi Multiple Campus

*In partial fulfillment of CACS 301 MIS & E-Business requirements for the fifth semester
of the Bachelor of Computer Applications program*

Submitted by:

Name: Nishan Sapkota

Roll No: 31

Name: Bikey Baniya

Roll No: 16

Under the Supervision of:

Name: Bishal Ghimire

Signature:



Tribhuvan University
Faculty of Humanities and Social Sciences
Mechi Multiple Campus

Supervisor's Recommendation

I hereby recommend that this report prepared under my supervision by **Nishan Sapkota and Bikey Baniya** entitled “**BN Organic**” in partial fulfillment of CACS301 MIS & E-Business requirements for the fifth semester of the Bachelor of Computer Applications program is recommended for the final evaluation.

.....

Mr. Bishal Ghimire

SUPERVISOR

Lecturer

Department of Computer Science and Applications

Mechi Multiple Campus

Bhadrapur, Jhapa



Tribhuvan University
Faculty of Humanities and Social Sciences
Mechi Multiple Campus

LETTER OF APPROVAL

This is to certify that this project prepared by **Nishan Sapkota** and **Bikey Baniya** entitled “**BN Organic**” in the partial fulfillment of the requirements for the degree of Bachelors in Computer Application has been evaluated. In our opinion, it is satisfactory in the scope and quality for the required degree.

..... Supervisor Bishal Ghimire Lecturer Department of Computer Science and Applications, Mechi Multiple Campus Bhadrapur-08, Jhapa.	 Program Deputy Director Raju Poudel Lecturer Department of Computer Science and Applications, Mechi Multiple Campus Bhadrapur-08, Jhapa.
..... Internal Examiner	 External Examiner

ABSTRACT

BN Organic Store is a haven for those seeking the freshest and most nutritious, sustainably grown produce. We offer a wide selection of organic fruits, vegetables, juices, and a range of natural products. We believe that access to wholesome food should be easy and enjoyable for everyone. At BN Organic Store, you'll find a vibrant selection of seasonal produce, from juicy berries and crisp apples to leafy greens and colorful root vegetables.

All our produce is sourced from local, organic farms, minimizing transportation distances and supporting local farmers. We prioritize sustainability and environmental responsibility in every aspect of our business. We strive to minimize our environmental impact by reducing plastic waste and promoting sustainable packaging options. We are committed to building strong relationships with our local farmers and supporting their efforts to cultivate healthy and sustainable food systems.

Our store also features a curated selection of organic pantry staples, superfoods, and natural health products, all carefully chosen to support a healthy and balanced lifestyle. You'll find a wide range of organic grains, nuts, seeds, and legumes, as well as a selection of high-quality supplements, teas, and natural sweeteners. We believe that healthy eating should be convenient and accessible, which is why we offer a variety of convenient and delicious options to make healthy choices easy.

We are committed to promoting healthy eating habits and fostering a community of health-conscious individuals. We regularly host workshops, cooking demonstrations, and community events to educate and inspire our customers on the benefits of organic living. Our knowledgeable staff is always available to answer questions and provide personalized recommendations to help you make informed choices about your food. We believe that by working together, we can create a healthier and more sustainable future for ourselves and our planet.

ACKNOWLEDGEMENT

We would like to express our special thanks of gratitude to our lecturer as well as our supervisor Mr. Bishal Ghimire for all his support and who gave us the golden opportunity to carry out this wonderful E-Commerce project on the topic 'BN Organic, which also helped us in doing a lot of research and we came to know about many things and ideologies regarding the topic.

We are overwhelmed in all humbleness and gratefulness to acknowledge our depth to all those who have helped us to put these ideas, well above the level of simplicity and into something concrete. We would also like to express our special thanks of gratitude to Mr. Raju Poudel, BCA Program Deputy Director, Mechi Multiple Campus (MMC) and all staff of MMC for their valuable support in doing this E-Commerce project.

Any attempt at any level cannot be satisfactorily completed without the support and guidance of our parents and friends. Therefore, we would like to thank our parents who helped us a lot in gathering different information and guiding from time to time in making this project despite of their busy schedule.

Nishan Sapkota

Bikey Baniya

Table of Contents

Chapter 1: Introduction.....	1
1.1. Introduction	1
1.2. Problem Statement	1
1.3. Objectives	1
1.4. Scope and Limitation	1
Chapter 2: Background Study and Literature Review.....	2
2.1. Background study	2
2.2. Literature Review.....	2
Chapter 3: System Analysis and Design	3
3.1. System Analysis.....	3
3.1.1. Requirement Specification.....	3
3.1.4. Process Modeling (DFD)	5
3.2. System Design	5
3.2.1. System Architecture	5
3.2.3. Interface Design (UI Interface/ Interface Structure Diagrams).....	6
Chapter 4: Implementation and Testing	7
4.1. Implementation	7
4.1.1. Tools Used	7
4.1.2. Modules and Plugins Used.....	7
4.2. Testing.....	8
4.2.1. Test cases for Unit Testing:	8
Chapter 5: Conclusion and Future Recommendation	10
5.1. Lesson Learnt.....	10
5.2. Conclusion.....	11
5.3. Future Recommendation	11
APPENDICES	12

List of Figures

Figure 1: Agile Methodology	3
Figure 2: Context Diagram	5
Figure 3: System Architecture	5
Figure 4: UI Interface	6

List of Tables

Table 1: Feasibility Analysis	4
Table 3: Test case for Add to Cart	8
Table 4: Test case for Checkout	9

ABBREVIATIONS

AGILE: Adaptive Growth through Innovative Leadership and Evolution

HTML: Hypertext Markup Language

CSS: Cascading Style Sheets

OMS: Order Management System

CMS: Content Management System

DXP: Digital Experience Platform

APIs: Application Programming Interfaces

Chapter 1: Introduction

1.1. Introduction

BN Organic is a store where you can find stylish and high-quality clothes and accessories. We prioritize sustainability by using eco-friendly materials like organic cotton and recycled fibers. We support brands that are committed to ethical production and minimizing their environmental impact. You'll find a diverse range of clothing options, from everyday essentials to special occasion outfits, all designed with your style and the environment in mind. We believe that fashion can be both stylish and sustainable. At BN Organic, you can find great clothes that look good and feel good knowing they were made with care for the planet. We invite you to explore our collection and experience the difference that sustainable fashion can make.

1.2. Problem Statement

The problems in most of the watch store are:

- a. Need to see fruits and vegetables physically to buy products.
- b. Difficult to select and search required products.
- c. More time consuming to buy and make payment for the products bought.

1.3. Objectives

- a. To ease the buying and selling of organic products by making them available on online platform.
- b. To ease the payment process by making online payment direct through bank or by using mobile banking platform like e-Sewa.

1.4. Scope and Limitation

By the end of this project, this project will benefit customer by saving their time and effort. In another words, this project will provide simplicity to customer to buy and make payment easily just by sitting on the couch of their own bedroom. This project will also be helpful for Sellers as they can upload their products details on the website and sell their products. This project is beneficial for both customers and sellers as it helps to establish good relationship among customers and sellers. The scope of this project is to reduce the time and effort of both customers and sellers, as there is the growth in technology, it allows us to add many new features in our website in future.

Chapter 2: Background Study and Literature Review

2.1. Background study

BN Organic is an e-commerce portal dedicated to providing high-quality organic groceries and eco-friendly products. It allows customers and sellers to connect, making it convenient to buy and sell organic food items and sustainable goods online. Organic groceries are a prominent category in the health and wellness market, featuring items such as fresh produce, whole grains, dairy alternatives, herbal teas, and natural snacks. These products are sourced from certified organic farms and crafted with a focus on sustainability, free from harmful chemicals and artificial additives, offering unique health benefits and a reduced environmental footprint. Organic groceries not only promote a healthier lifestyle but also contribute to environmental preservation, embodying wellness and sustainability. As the market continues to grow, manufacturers and retailers must adapt to changing consumer preferences and emerging trends to remain competitive.

2.2. Literature Review

BN Organic is Nepal's premier online shopping destination for organic groceries and eco-friendly products. As part of the ATC Group, it aims to create a reliable and seamless e-commerce ecosystem, offering life-enhancing experiences for buyers and sellers alike. With over 5,000 products in its growing inventory, BN Organic is committed to providing a wide range of high-quality organic food items and sustainable goods that cater to the health-conscious and environmentally aware consumer.

Dedicated to customer satisfaction, BN Organic ensures top-quality products at competitive prices while upholding the highest standards of integrity and service. Each member of the sales team is extensively trained in the products offered, delivering knowledgeable support to enhance the shopping experience. Customers can easily browse BNOrganic.com to explore an extensive selection of organic groceries and eco-friendly products, with orders delivered conveniently to their doorstep.

BN Organic embodies the principles of health, sustainability, and convenience, making organic living accessible to everyone. By continuously adapting to changing consumer preferences and emerging trends, BN Organic strives to remain a leader in Nepal's e-commerce industry, promoting a healthier lifestyle and a greener planet.

Chapter 3: System Analysis and Design

3.1. System Analysis

Since, the system initial requirements are well understood and the project deadline is also fixed already and there may be the chance of changes in technology and some changes must be made in the system in future. The project teams believe that the best approach in this environment is following iterative nature. Each iteration will contain set of SDLC steps (planning, Analysis, Design, Implementation). Agile Methodology allows for regular testing to see whether the product is working or not during development phase. It demonstrates the working functionalities. It gives freedom when recent changes are needed to be implemented. Agile means faster product release since less time is there to complete the project So, this is appropriate method of software development. [4]

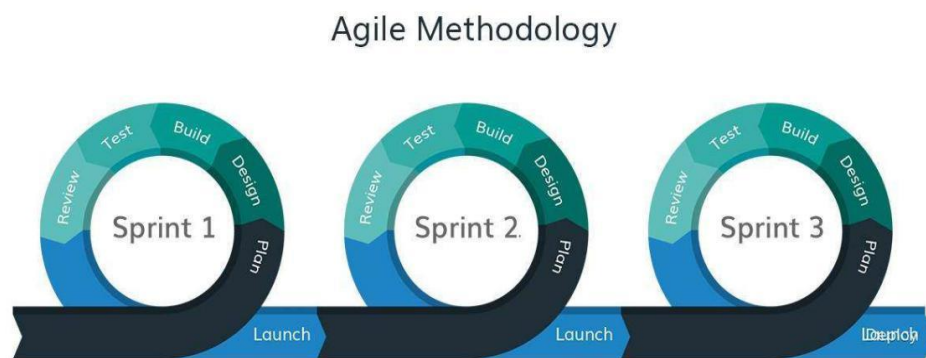


Figure 1: Agile Methodology

3.1.1. Requirement Specification

The software requirements are description of features and functionality of the target system. Requirements convey the expectations of users from the software product. The requirements can be obvious or hidden, known or unknown, expected or unexpected from client's point of view. So, it's important to make them clear both for the development team and the stakeholders. The known requirements of the system to be developed are divided into two categories (functional and non-functional) and are mentioned below:

i. Functional Requirements

1. System should allow Customer to order the Product.
2. System should allow admin to add, update, and delete product details.
3. System should allow to view the product details.
4. Customer can search the products.
5. Customer should be able to make payment through the website.
6. Customer Order Details report should be generated.
7. Customer can review the products.
8. Customer can give feedback the products.

ii. Non Functional Requirements

Reliability Requirement

The system must perform accordingly to the member request. Transactions must be tracked efficiently. Data and information should not be lost. Changes must be reflected to the database. Besides, the registration should validate for wrong input of information.

Usability Requirements

The system must be designed with user-friendly and easy to use by the staff so that the user can perform their job nicely. It must have a clear instruction to guide user through the system. Besides that, the description of error message should be clear.

Implementation Requirements

While implementing the system, XAMPP must be installed. When we launch the system online, users can directly order from our portal.

Security Requirements

The system must be highly secured. User records manipulation is only accessible to the admin level user.

3.1.2. Feasibility Study

A feasibility study is a way to evaluate the practicality and desirability of a project. Before a company invests time and money into a project, they need to know how successful the project will be before investing. Sometimes companies want to understand input costs, the amount of research that will need to be done, or even the marketability of a project.

Table 1: Feasibility Analysis

Name	Description	Remarks
Technical Feasibility	It is developed under MacOS operating system using WordPress.	Therefore, considering the requirement of the system is easily accessible the system is technologically feasible.
Operational feasibility	the system is designed under Content Management System	As it is user-friendly designed, the system is operationally feasible.
Economic Feasibility	The system is built using free software technologies and there is no any recurring cost just internet connection is needed.	Since open source technologies are used and there is no extra cost needed, the system is economically feasible.

3.1.4. Process Modeling (DFD)

Context Diagram

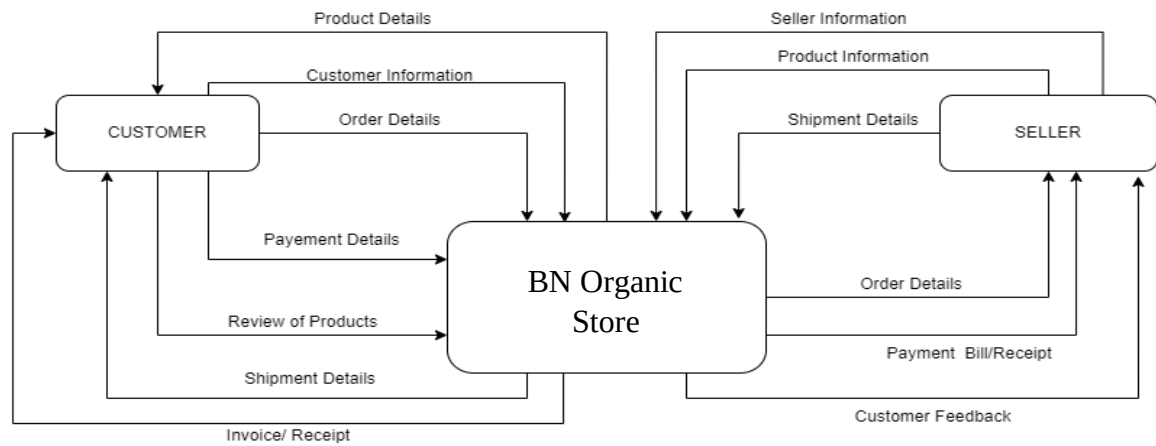


Figure 2: Context Diagram

3.2. System Design

3.2.1. System Architecture

A System architecture diagram also known as block diagram is a diagram of a system in which the principal parts or functions are represented by blocks connected by lines that show the relationships of the blocks. The block diagrams below show a simple visual representation for describing actions of the complex system. The blocks are often referred as black boxes, which represent mathematical or logical operations that occur in sequence. The base architecture of the system to be developed is shown below where basic pictorial representation has been drawn to narrate the overview of the system.

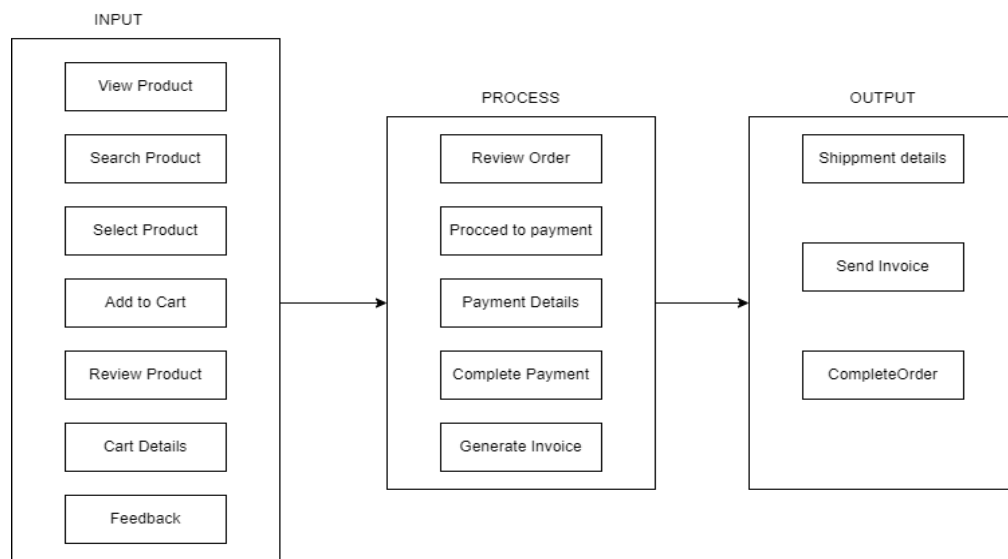


Figure 3: System Architecture

3.2.3. Interface Design (UI Interface/ Interface Structure Diagrams)

The user interface for this software will start from a welcome splash screen. Users will be prompted to the respective dashboard. The dashboard are designed in such a way that they can easily manage and manipulate the data and information they have access to. Furthermore, they will get interfaces accordingly depending on what action they perform. Tool-tips are placed in ambiguous component so that users will get better experience. Since Customer gets access to minimum actions, the interface is also organized in a way that user can easily access actions via menu bar. Samples for UI (Screenshots) are shown below at Appendices.

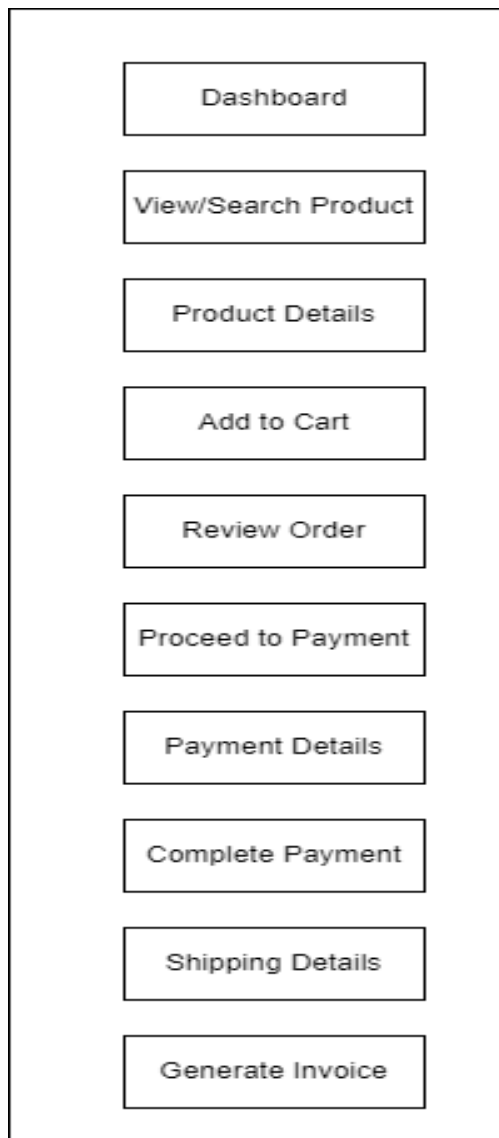


Figure 4: UI Interface

Chapter 4: Implementation and Testing

4.1. Implementation

4.1.1. Tools Used

Content management system

Content management systems (CMS) give e-commerce businesses the ability to create, edit, and publish the content on their website. With a CMS you can customize your website presentation by modifying the layout of your product pages, homepage, or any other part of the online store. With a modular architecture, the CMS is headless. Instead of being attached to the e-commerce platform or other backend components, it simply makes content available via RESTful APIs. This is essential to creating memorable customer experiences. The headless CMS allows you to create different frontend experiences for different use cases. Brands and retailers that want to create unique content across multiple channels can use a digital experience platform (DXP) for their content management module. A DXP comes with a headless CMS built-in. It also includes other components such as data and analytics tools, A/B testing software, and user data tools. This allows marketers to add a high level of personalization to content creation. With a DXP like Fabric XM, you are not limited to your own e-commerce site. Through APIs, you can deliver content to mobile apps, IoT devices, or any other sales channel.

4.1.2. Modules and Plugins Used

Product information management

The product information management (PIM) module contains all the information related to the products in the e-commerce catalog. Some of the information managed by PIM includes:

- Title
- Descriptions
- Attributes
- Technical specification
- Images

PIM also handles the taxonomies and relationships between your products. This allows you to segment the display of products, making it easier for customers to search for the items they are looking for. PIM acts as a single source of truth. You can upload data from any number of sources. Product details can then be spread to any channel. A single hub for product data helps ensure all information is up to date and free from errors and discrepancies. The software connects to other applications through APIs. This allows the other parts of your e-commerce system to search and retrieve important product data.

Shopping cart and checkout

The shopping cart allows customers to select and hold the products they wish to purchase. It tracks the user's session, allowing them to leave the site and come back later with items still in their cart. During the checkout process, the cart collects the customer's payment information. This information is passed to the third-party payment processor. The details of the order are sent to other modules like the order management system (OMS), inventory management system, and customer relationship management (CRM) system. The shopping cart has a database to store and retrieve the data necessary for the checkout process. There are three types of data including static, session, and processed data. Some order management systems like Fabric OMS include cart and checkout functionality. With cart and order APIs, you can process orders from any sales channel.

Plugins Used

e-Sewa: eSewa – Nepal First Payment Gateway plugin enables any user with a wordpress site to integrate eSewa payment gateway in their website/web application. eSewa is Nepal's first online payment gateway and is a comprehensive online payment solution. It is widely popular, secure as well as one of the most trusted online payment gateway. The following plugin incorporates itself with WordPress woo-commerce plugin and lets users pay via their eSewa id to buy any virtual goods or services.

4.2. Testing

Unit and integration testing is done in each module. In Unit testing small testable parts of the application are tested individually and independently. The team members themselves did unit test and the test was used for database test, records of each table, basic function test, add student, add book, add member.

In integration testing, each module is combined and tested as a group. This test verified that the system maintained data integrity and can operate in coordination with other systems in the same environment. The team members themselves carried out the test.

4.2.1. Test cases for Unit Testing:

a. Test case for Add to Cart

Table 2: Test case for Add to Cart

S.N	Test Case	Expected Result	Remarks
1	Enter the number of products to the cart equal to 0	System should display message the value must be greater than 0.	Pass
2	Enter the number of products greater than 0.	System should add the number of product entered into the cart.	Pass

b. Test case for Checkout

Table 3 Test case for Checkout

S.N	Test Case	Description	Expected result	Remarks
1.	Enter Billing First & Last Name	Enter the name of the customer.	System should enter the name of the customer.	Pass
2.	Select the Country/Region	Select the respective country or region	System should select the respective country.	Pass
3.	Enter the Street Address	Enter the street address of the customer	System should enter the customer address.	Pass
4.	Searching products	Enter product name	System Should display the searched product.	Pass
5.	Enter phone number	Enter the 10 number length phone number	System should enter the phone number.	Pass
6	Enter the Email address	Enter the proper format email address	System should enter the email address.	Pass

Chapter 5: Conclusion and Future Recommendation

5.1. Lesson Learnt

The BN Organic e-commerce report highlights several crucial lessons for businesses in the organic groceries and sustainable products industry.

The importance of a user-friendly website: The report found that customers are more likely to buy products from an e-commerce platform that is easy to navigate and offers a seamless browsing experience. This emphasizes the need for investing in a well-designed website tailored to the needs of shoppers, making the buying process smooth and enjoyable.

The importance of product reviews: Customers heavily rely on product reviews when deciding on purchases. The report underscores the significance of encouraging shoppers to leave reviews and prominently showcasing positive feedback on the platform. Honest reviews build trust and can greatly influence buying decisions in the competitive fashion and accessories market.

The importance of product images: High-quality visuals play a vital role in driving sales. The report revealed that customers are more inclined to purchase products with professional and detailed images. This highlights the necessity of investing in top-notch product photography to accurately represent apparel and accessories, making them appealing to potential buyers.

The importance of mobile optimization: With the growing use of smartphones for online shopping, the report stresses the need for an e-commerce platform optimized for mobile devices. A seamless mobile experience ensures that customers can effortlessly browse and make purchases, catering to the on-the-go shopping behavior of modern consumers.

The importance of customer service: Responsive and helpful customer service remains a top priority for customers. The report emphasizes the value of investing in efficient customer support systems, ensuring that shoppers can quickly resolve inquiries or concerns, fostering loyalty and satisfaction.

Overall, the report highlights the importance of creating a user-friendly website, promoting product reviews, leveraging high-quality images, optimizing for mobile use, and providing exceptional customer service to succeed in the competitive BN Organic groceries and products market.

5.2. Conclusion

The BN Organic project aimed to develop a system that would streamline the process of buying and selling organic groceries and eco-friendly products through an online marketplace. Through the implementation of this system, we have achieved a more efficient and organized approach to shopping for organic products, resulting in time savings and improved customer satisfaction.

The BN Organic e-commerce portal is a valuable tool for managing and purchasing organic groceries. It offers a user-friendly interface, a wide range of high-quality and sustainable products, and efficient ordering and delivery processes. The system also provides valuable insights into customer preferences and emerging trends in organic living, which can help businesses better understand their customers and make informed decisions.

Overall, the BN Organic e-commerce portal is an essential tool for anyone looking to manage and purchase organic groceries and eco-friendly products conveniently and effectively.

5.3. Future Recommendation

Based on the project, we recommend that the BN Organic e-commerce platform continue to focus on expanding its product offerings to include a wider variety of high-quality, organic, and eco-friendly products. This could include introducing sustainably sourced specialty items, seasonal produce, and customizable bundles to cater to diverse customer preferences. In addition, we recommend that the platform enhance the customer experience by providing detailed product descriptions, nutritional information, user reviews, and live customer support. Adding interactive features such as meal planning tools or product comparison options could further improve the shopping experience.

Furthermore, we suggest focusing on building a strong and engaging online presence through targeted social media campaigns, influencer collaborations, and content marketing tailored to eco-conscious consumers. Search engine optimization and paid advertising can also help increase visibility and attract more traffic to the website. Offering exclusive deals, memberships, or loyalty programs for regular customers can encourage repeat purchases and foster brand loyalty.

Finally, the platform should aim to provide a seamless shopping experience by streamlining the checkout process, offering multiple payment options, and ensuring transparency in shipping, packaging, and return policies. By prioritizing customer satisfaction and addressing their unique needs, the e-commerce platform can establish itself as a trusted and go-to destination for organic groceries and eco-friendly products.

APPENDICES

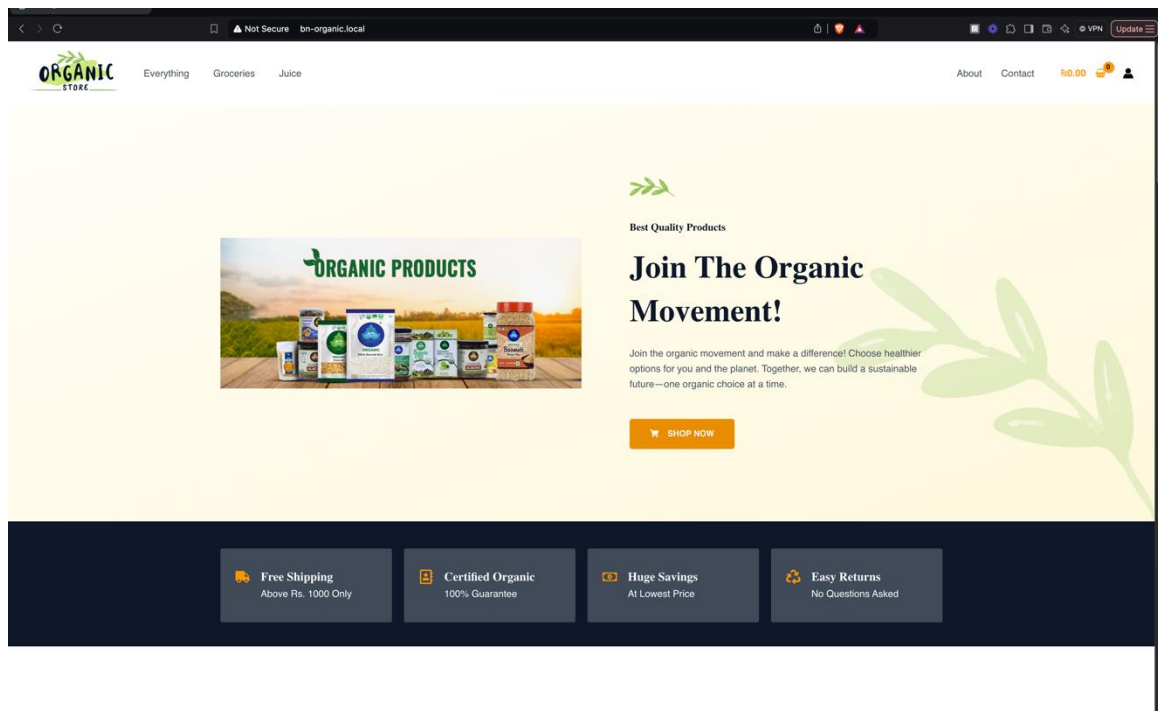


Fig:- Dashboard

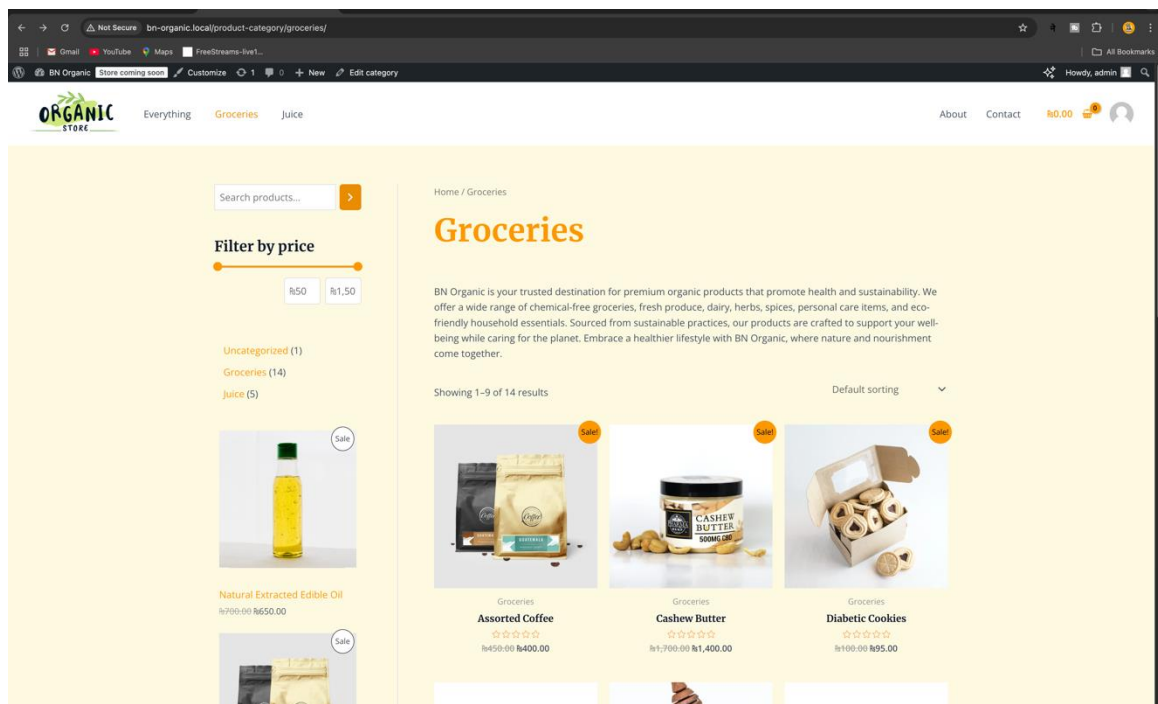


Fig:- Products page

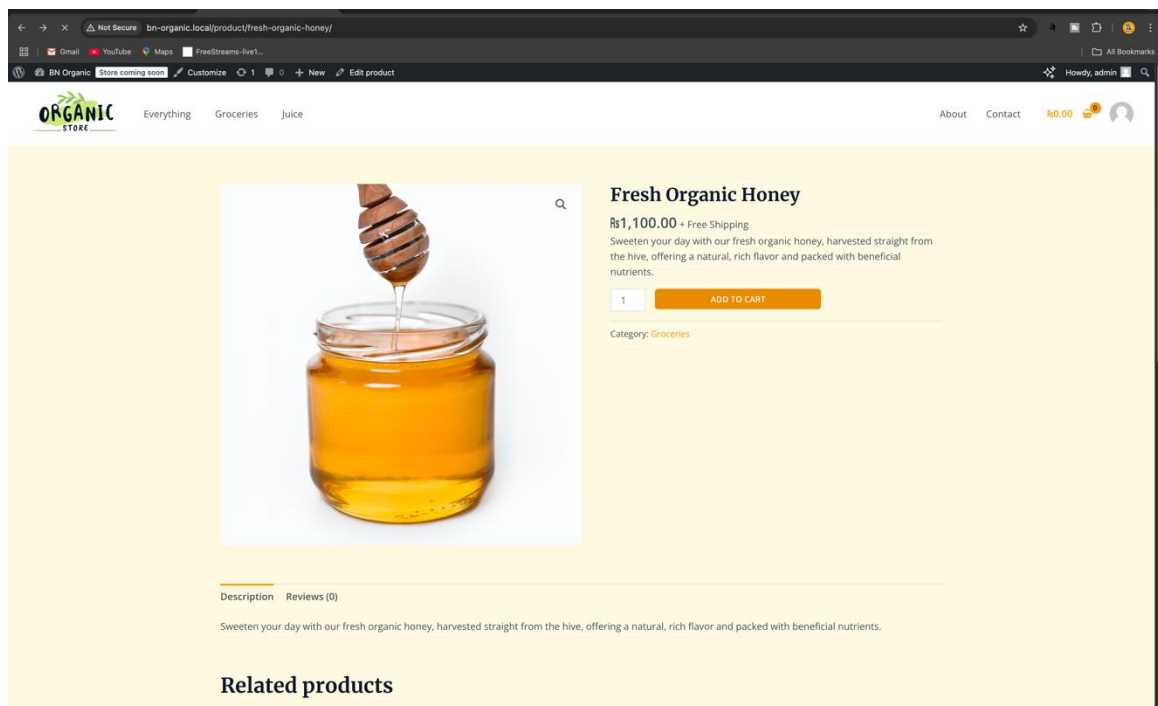


Fig:- Product Details

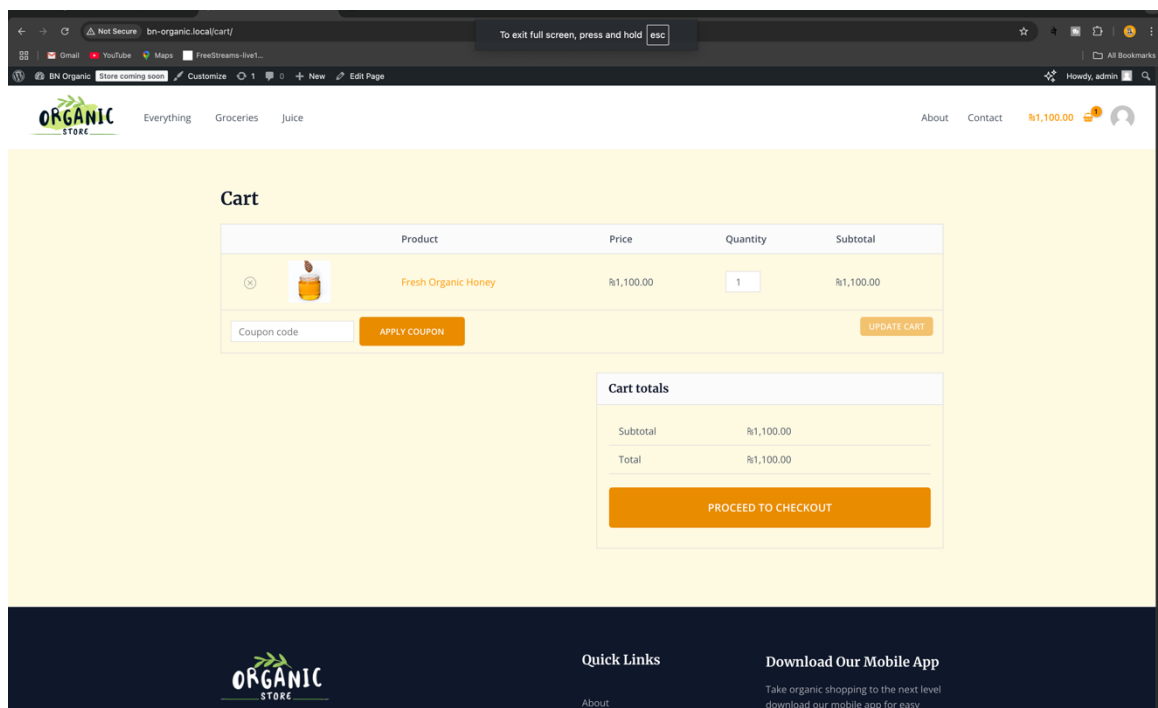


Fig:- Add to Cart

The screenshot shows a web browser window with the URL `bn-organic.local/checkout/`. The page is divided into two main sections: "Billing details" and "Your order".

Billing details:

- First name ***: Nishan
- Last name ***: Sapkota
- Company name (optional)**: (empty field)
- Country / Region ***: Nepal (dropdown menu)
- Street address ***: Jhapa
- Town / City ***: Bhadrapur
- State / Zone ***: Mechi (dropdown menu)
- Postcode / ZIP (optional)**: 57200
- Phone ***: 9853211043
- Email address ***: dev-email@wpengine.local
- Additional information**: (empty field)

Your order:

Product	Subtotal
Fresh Organic Honey × 1	₹1,100.00
Subtotal	₹1,100.00
Total	₹1,100.00

Below the order summary, there is an eSewa logo and a message: "Pay via eSewa, Securely and Reliably SANDBOX ENABLED. You can use testing accounts only." A note states: "Your personal data will be used to process your order, support your experience throughout this website, and for other purposes described in our [Privacy policy](#)." At the bottom of this section is an orange button labeled "PROCEED TO ESEWA".

Fig:- Billing address and Delivery Address

The screenshot shows the E-Sewa interface. At the top, there is a navigation bar with the "eSewa" logo and a language dropdown set to "English".

The main content area is split into two panels:

- Left Panel (Payment Summary):**
 - EPAYTEST** logo.
 - Total Amount**: NPR. 1,100.00
 - A promotional banner for "FEEL GOOD with Great Food" featuring a bowl of food and a "Flat 15% Discount" tag.
- Right Panel (User Interaction):**
 - Sign in to your account** header.
 - Input fields for **eSewa ID** and **Password/MPIN**.
 - A checkbox for **I'm not a robot** with a CAPTCHA image.
 - A green **LOGIN** button.
 - A link for **Forgot Password?**.
 - A link for **Don't have an account? Register**.
 - A **CANCEL PAYMENT** link at the bottom.

At the bottom of the page, there is a footer with the text "©eSewa Nepal. 2025. All Rights Reserved." and links for [Terms & Conditions](#) and [Privacy Policies](#).

Fig:- E-Sewa Interface